Active Six-DOF Magnetic Levitator for Fully Automated High-Throughput Large-Range Single-Mode Optical Fiber Alignment

Hyeong Min Yoon, Chang Hwan Kim, Kyujin Kang, Jong Min Sung, Eun Kyu Kim, Jae Woo Jung, JungMin Park, and Jun Young Yoon

***Abstract*—This article presents a compact hybrid-reluctance-type six-degrees-of-freedom (DOF) magnetic levitator combined with a novel dynamic spiral searching (DSS) algorithm for fully automated high-throughput alignment of single-mode optical fibers. Conventional alignment methods rely on stacked actuator stages, resulting in complexity, increased footprint, and limited throughput. In contrast, the proposed system achieves long-range motion, high-speed response, and sub-micrometer precision within a single, integrated actuator stage actively controlled in all six degrees of freedom. The DSS algorithm adaptively modulates both the amplitude and the center position of the spiral trajectory based on real-time optical coupling feedback, eliminating the need for the conventional first-light detection, and thus enabling rapid alignment from any arbitrary fiber misalignments. Experimental results demonstrate sub-micrometer tracking accuracy and complete alignment times under 0.9 s, which is up to 86.4% faster than conventional hill climbing methods, with robust performance across randomly positioned optical fibers. This unified framework simplifies the overall actuator architecture, providing a practical solution for photonics packaging that requires high-speed, high-range, and high-precision alignment.**

***Index Terms*—Hybrid-reluctance-type actuation, magnetic levitation, multi-degrees-of-freedom (DOF) control, optical fiber alignment.**

I. Introduction

Coupling different optical fibers is an essential process in opto-electronics packaging [1]. Historically, such task required significant time and manual effort; however, automation has dramatically improved efficiency and throughput. During the coupling process, transmission loss is unavoidable due to the fiber misalignment, which is mainly caused by axial deviations [2], [3]. Hence, precise fiber-to-fiber or laser-to-fiber alignment is a crucial process in coupling automation. This process is, however, highly challenging as it requires accurate positioning of the optical fibers with microscopic dimensions across multiple degrees-of-freedom (DOFs). The core size of a typical single-mode fiber (SMF) is 9 μm in diameter and can be as low as 5 μm for the single-mode microstructured optical fibers (MOFs) [4], [5]. Achieving a precise alignment at the sub-micron level is essential to ensure a satisfactory coupling loss level, which should remain below 1% of the peak transmission intensity [6].

The automation of the optical fiber alignment can be classified into two categories: passive and active methods. Passive alignment methods are advantageous due to their simplicity, as they do not require feedback [7]. However, their alignment precision heavily depends on the fabrication tolerances of the key alignment structures, and the accuracy is reported to be 1–2 μm, which is significantly below the required sub-micron level [8], [9]. In contrast, the majority of the precise fiber alignment modules nowadays employ active alignment methods with closed-loop operation of the actuator modules, resulting in sub-micron alignment accuracy [10]. The feedback signal is typically the coupled power measured by an optical power meter or the relative position of both optical fibers determined by a camera-based vision system [11], [12].

A typical active optical fiber alignment process involves coarse and fine alignment [13]. The main goal of the coarse alignment step is to catch the so-called "first light," which is the approximate optimum position. Spiral, square, or raster scanning schemes are the most widely used for the coarse alignment [14]. For such process, the actuator module should be capable of moving over a large range, typically a few hundred micrometers of translational motion [15]. In addition, the actuation speed needs to be sufficiently fast, as the coarse scanning generally takes up a large portion of the total alignment time [16]. After finding the first light, the position is finely tuned to find the exact optimum point with the maximum transmission power. The hill climbing algorithms are widely adopted for the fine alignment process in the current photonics packaging industry due to its simple structure and the effective performance [17].

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In such process, precise positioning of the actuator module is crucial for the alignment accuracy performance.

The prevalent actuator types for the optical fiber alignment are piezoelectric actuators and motorized hexapods. Piezoelectric actuators offer a fast response time and precise positioning at the sub-micrometer level. However, their range of motion is limited to a few tens of micrometers [18], [19]. Such limitation makes them highly dependent on the initial positioning. As a result, they are not solely used in automated processes but are frequently employed in experimental research where a manual coarse alignment is feasible [20]. In contrast, hexapods excel in the coarse alignment due to their larger range of motion. However, they have drawbacks in fine-tuning due to the slow response time and limited precision caused by backlash and friction [21], [22]. Increasing demands for the optical fiber alignment in clean vacuum conditions pose a serious challenge for hexapods, as friction and mechanical bearings can generate particles unsuitable for vacuum environments [23]. Piezoelectric actuators are often stacked on hexapods to leverage the strengths of both types, but such architecture significantly increases the complexity and footprint of the overall alignment system [24], [25]. Furthermore, it suffers from limited throughput as each stage must be operated sequentially.

Magnetic levitation systems have been actively utilized in various industrial fields owing to their strength in contact-free operation without bearings or flexures. However, they have not been applied to optical fiber alignment, because force-generation units for multi-DOF control are physically separated and therefore configured in a bulky way requiring a large footprint, which limits their applicability to fiber alignment tasks [26]. While Lorentz-type magnetic levitators can move over a long range, they have relatively low force constants, limiting the quick response capabilities [27]. Reluctance-type levitators, in contrast, offer a large force constant but suffer from low precision due to the nonlinearity [28]. To address such challenges, this article presents a compact hybrid-reluctance-type magnetic levitator design for the rapid and precise alignment automation, utilizing two distinct magnetic flux sources from permanent magnets (PMs) and coil windings. The contributions of this article include: 1) the proposal of a new compact hybrid-reluctance-type six-DOF magnetic levitator design that enables large-range motion and sub-micrometer precision suitable for optical fiber alignment applications; 2) capability validation of the proposed active six-DOF magnetic levitator in achieving the rapid and accurate optical fiber alignment, suggesting its strong potential in the high-performance optical fiber alignment systems; and 3) the development of a simple, practical, yet highly effective automation framework that exploits the intrinsic actuation characteristics of the hardware system. The remainder of this article is organized as follows. Section II presents the magnetic design and working principle of the proposed magnetic levitator, followed by the detailed configuration of the prototype system in Section III. Section IV then discusses the active stabilization method and motion tracking strategies for optical fiber alignment. Section V proposes a novel dynamic spiral searching (DSS) algorithm for fully automated high-throughput alignment and experimentally validates the single-mode optical fiber alignment performance. Section VI then concludes the article.

II. Design of Active Six-DOF Magnetic Levitator

Fig. 1 shows a computer-aided design (CAD) model of the proposed six-DOF magnetic levitator designed for high-throughput, high-precision, and large-range optical fiber alignments. The overall assembled dimensions of the proposed levitator are $134 \times 140 \times 70$ mm. For comparison, the commercial hexapod for fiber alignment occupies approximately $136 \times 136 \times 126$ mm, and the piezo nanopositioner typically stacked on top of the hexapod adds another $40 \times 40 \times 40$ mm [29], [30]. Commercial stacked alignment systems therefore combine these two layers in series, whereas the proposed levitator achieves full six-DOF actuation in a single, more compact module [31]. The stationary part of the module consists of the upper and lower stators, each shaped as an "H," and a center stator shaped as a pound sign (#). The rectangular-shaped levitated rotor carries an optical fiber and is actively controlled in all six DOFs. PMs are positioned between the stators, generating a biased flux through the airgaps between the rotor and the stators. This section presents the electromagnetic design and the operating principles of the proposed six-DOF magnetic levitator module.

A. Electromagnetic Design

Fig. 2 provides schematic cross-sectional top views of the six-DOF magnetic levitator. The rectangular rotor forms a total of 16 airgaps with the stators: eight airgaps in the x-y plane against the center stator along the red dashed lines as shown in Fig. 2(a) and the other eight in the z direction against the upper and lower stators as indicated with red circles in Fig. 2(a) and (b). To generate the PM-biased fluxes necessary for the hybrid-reluctance actuation in six DOFs, two PMs magnetized in the z direction are located between the stators, directing the PM-biased fluxes (red dashed lines) from the center stator to the rotor through the airgaps as shown in Fig. 2(a). The return paths for such biased fluxes are formed through the upper and lower stators as illustrated in Fig. 2(b). These biased fluxes generate reluctance forces between the stators and rotor in all 16 airgaps, which inherently exhibit negative stiffness. By superimposing current-driven fluxes on top of such PM-biased fluxes, however, the magnetic flux distribution in the airgaps can be modulated so as to generate net forces and torques for the rotor to be actively and stably controlled in all six DOFs. The force and torque generation mechanism is discussed in the following section.

B. Six-DOF Forces and Torques Generation

There are a total of 16 coil windings wound around the stator cores: eight coils on the center stator and the other eight coils on the upper and the lower stators. When current flows through these actuating coils, it induces a magneto-motive force, forming closed current-driven flux paths. The reluctance of the airgaps between the stators and the rotor plays a dominant role in determining the magnetic flux path, as the flux naturally follows

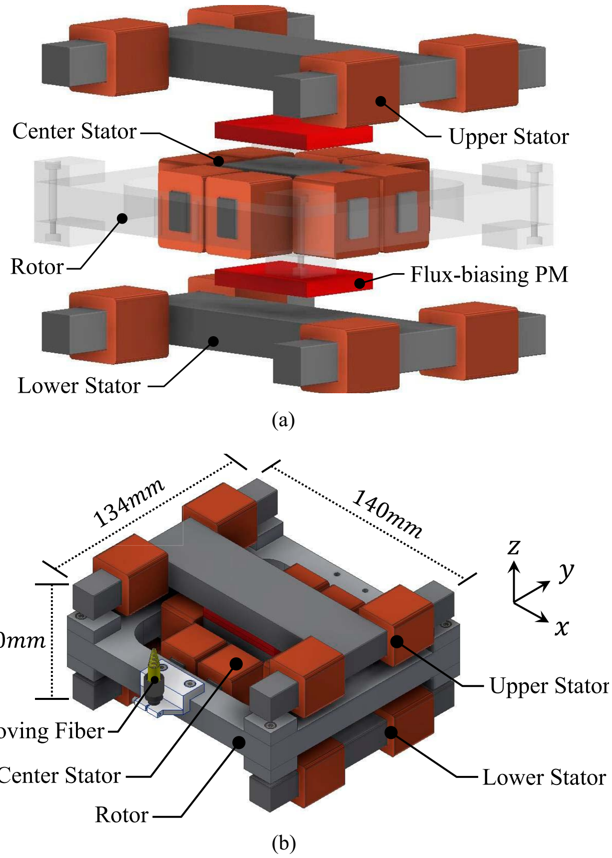


Fig. 1. 3-D rendering of the proposed six-DOF magnetic levitator. (a) Exploded view showing three parts of stators (upper, lower, and center) connected by flux-biasing PMs and a rectangular rotor. (b) Assembled view with a moving fiber carried by the rotor.

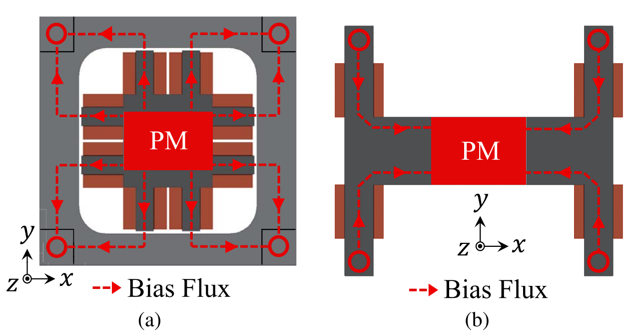


Fig. 2. PM-biased magnetic flux distribution in the proposed six-DOF magnetic levitator. (a) Cross-sectional view of the center stator and the rotor showing eight airgaps in the x-y plane. (b) Cross-sectional view of the upper or the lower stator showing four airgaps in the z direction (denoted as red circles), thus a total of eight airgaps for the upper and lower stators. Note that the flux-biasing PMs are magnetized in the z direction.

the path of least reluctance. This unbalanced flux distribution in the airgaps results in net forces or torques being applied to the rotor, enabling control of all six DOFs.

Fig. 3 illustrates specifically how the superposition of the current-driven flux (blue lines) and the PM-biased flux (red dashed lines) generates the six-DOF forces and torques. For instance, in Fig. 3(a)–(c), the imbalance in flux density around the center stator generates in-plane net forces and torques. Similarly, the out-of-plane forces and torques are generated as shown in Fig. 3(d)–(f), enabling active control in all six DOFs. Furthermore, the flux superposition principle illustrated in Fig. 3 decouples the multi-input multi-output (MIMO) system into six independent single-input single-output (SISO) systems at the hardware level, and each DOF is actuated by a distinct current combination without coupling to the others. This hardware-level decoupling allows each DOF to be analyzed and verified independently, and we experimentally validate the resulting cross-axis decoupling in Section IV-B. As a result, this decoupling enables: 1) fast servo control in the x and y directions for axial alignment of optical fibers; 2) precise distance regulation in the z direction; and 3) active magnetic bearing support for rotational DOFs.

TABLE I
KEY DESIGN PARAMETERS OF OUR MAGNETIC LEVITATOR MODULE

Parameter	Description	Value
m	Rotor mass	1.29 kg
I_x	Rotor x-axis moment of inertia	3704.5 kg · mm^2
I_y	Rotor y-axis moment of inertia	3062.9 kg · mm^2
I_z	Rotor z-axis moment of inertia	6678.5 kg · mm^2
g_l	Lateral airgap length	1.5 mm
g_v	Vertical airgap length	4.5 mm
N_l	Center stator winding turns	240
N_v	Upper and lower stator winding turns	215
d	Coil winding wire diameter	0.7 mm

III. PROTOTYPE CONSTRUCTION

Given the decoupled six-DOF architecture described in Section II-B, the levitator is designed on a per-DOF basis, as summarized in the flowchart in Fig. 4. The flowchart outlines the procedure used to determine the key design parameters of the proposed six-DOF magnetic levitator prototype. To systematically analyze the system, the stiffness and force constant values are calculated through finite element analysis (FEA) simulations. Table I shows the preoptimized parameters selected for large-range motion and sub-micrometer precision within a compact footprint compared to conventional piezo-hexapod stacked systems. As a result, Fig. 5 shows an experimental prototype setup for demonstrating SMF alignment. The rest of the section describes the design and construction details of the rotor and the stators with the sensing system for active position control. Note that the core design process discussed in this section can be applied to more complex and diverse packaging scenarios without fundamental architectural changes.

A. Rotor and Stators

Fig. 6(a) shows the fabricated rotor of our optical fiber alignment module. The rotor is made up of 0.35 mm thick laminated steel plates in a rectangular shape, along with steel blocks positioned at the four corners. It weighs 1.29 kg and measures 144×134 mm^2, with a total height of 31 mm including the steel blocks. Using the rotor geometry and material properties, we

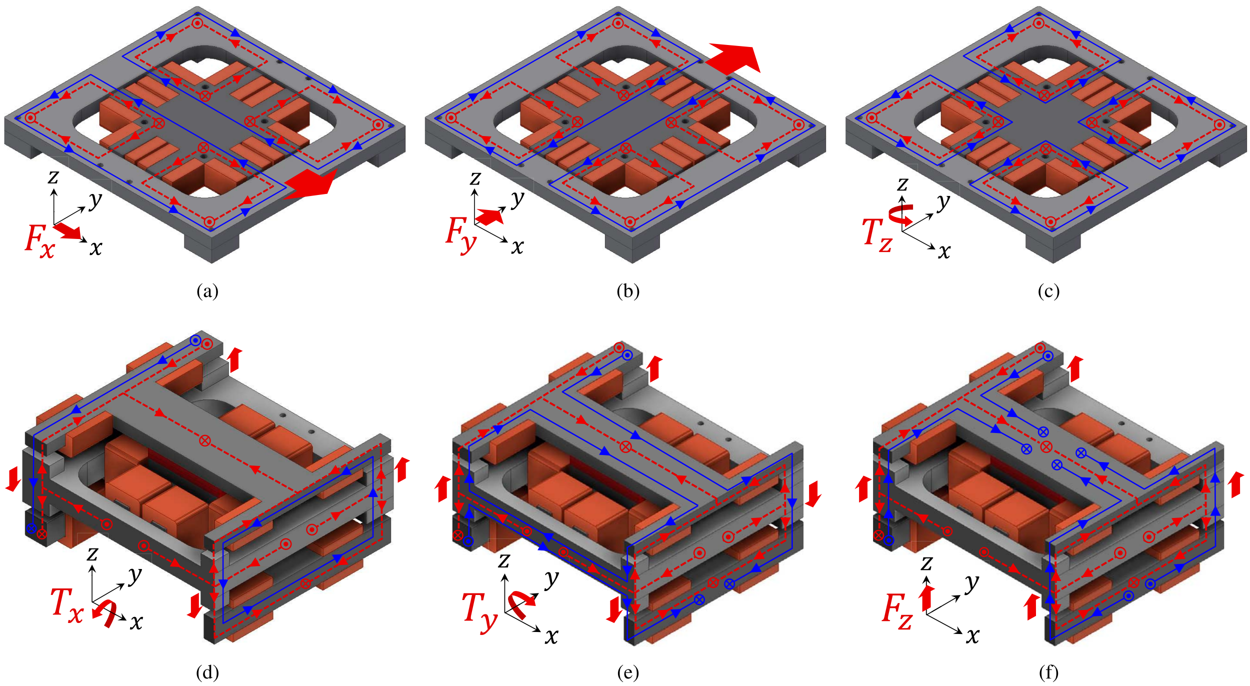


Fig. 3. Six-DOF force and torque generation mechanism in the proposed magnetic levitator. The PM-biased fluxes are represented with red dashed lines, and the current-driven fluxes are indicated by blue lines. By superposing the magnetic fluxes from two sources, the resultant fluxes in some airgaps are strengthened while weakened in the other airgaps, generating the net forces F and torques T in all six DOFs. Cross-sectional diagrams of the center stator showing the generation of (a) F_x, (b) F_y, and (c) T_z for the in-plane motions, and the diagrams illustrating the generation of (d) T_x, (e) T_y, and (f) F_z for the out-of-plane motions.

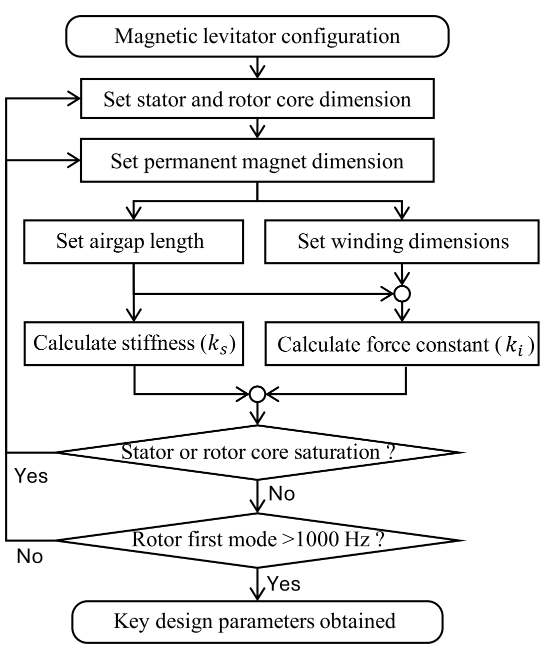


Fig. 4. Flowchart illustrating the procedure for determining the key design parameters for the proposed six-DOF magnetic levitator prototype.

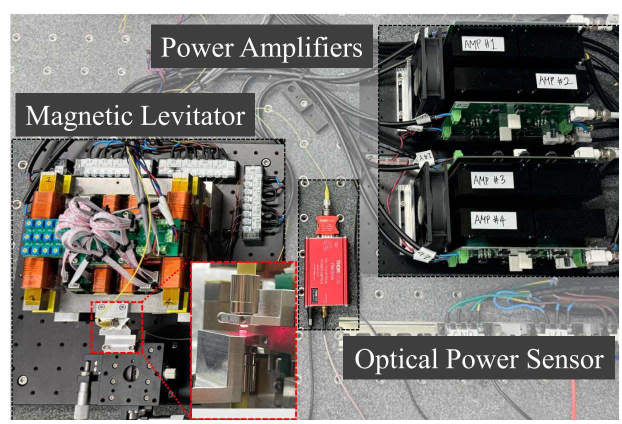
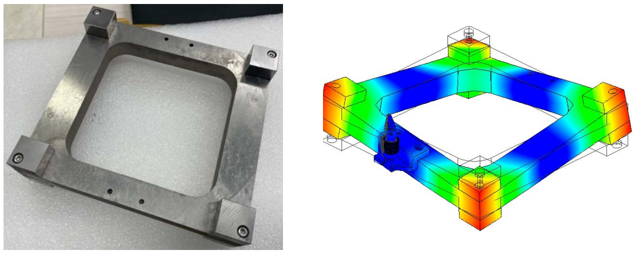


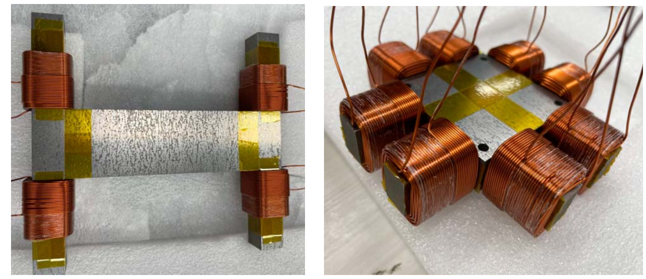
Fig. 5. Photograph of the magnetic levitator prototype for fully automated high-throughput optical fiber alignment. An 1 mW light is emitted from the moving fiber carried by the rotor and transmitted to the stationary fiber. The coupling efficiency between the two optical fibers is evaluated using an optical power sensor (S150C from Thorlabs). Note that the micro-stages are connected to the stationary fiber for deliberate misalignment.

calculate its structural modes using mechanical finite element method (FEM) modal analysis. Fig. 6(b) shows the first mode shape of the rotor, simulated at 1205 Hz with an optical fiber attached. This value is more than one order of magnitude higher than the controller crossover frequency used for active stabilization, and over two orders of magnitude higher than the major frequencies involved in the fiber alignment process. Therefore, it does not compromise the system stability nor significantly affect the alignment dynamics. Moreover, the mode shape is symmetrical around the center of mass of the rotor, facilitating collocated measurement with the sensor installation described in the following section.



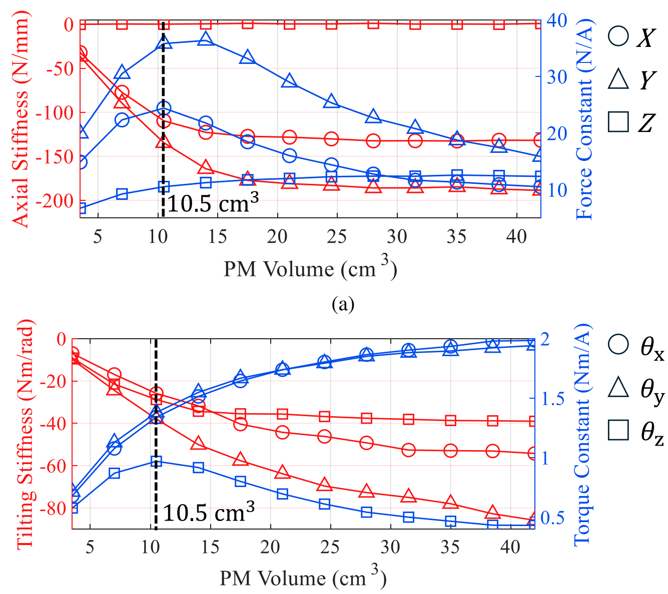
(a) (b)

Fig. 6. (a) Photograph of fabricated rotor before the installation with the stators. (b) First structural mode shape of the rotor at 1205 Hz calculated from an FEM modal analysis.



(a) (b)

Fig. 7. Photographs of prototype stators. (a) Upper and lower stators. (b) Center stator. The coil windings are inserted and fixed to the stator cores. Kapton tapes are wrapped around the stator cores for electrical insulation.



(b)

Fig. 8. Simulated stiffness (red lines) and actuation force (blue lines) characteristics with varying PM volumes. (a) Axial stiffness and force constant for X- (circle markers), Y- (triangle markers), and Z- (square markers) DOFs. (b) Tilting stiffness and torque constant for θ_x- (circle markers), θ_y- (triangle markers), and θ_z- (square markers) DOFs. The selected PM volume of 10.5 cm^3 used in the prototype is indicated by the black dashed lines.

The stators are composed of coil-inserted steel cores as shown in Fig. 7. The steel cores are also laminated and cut by wire-electrical discharge machining for the desired shape. The coil windings are made of 0.7 mm diameter copper wire. The wire diameter is selected to prevent excessive heat generation from the driving current while ensuring that the coil windings do not occupy excessive space. Although a larger number of winding turns increases the force constant, it also requires more space. Conversely, using thinner wires allows for more turns but results in higher current density and heat generation. Considering such tradeoff, the numbers of winding turns are selected as 240 for the coils inserted in the center stator and 215 for the coils located in the upper and lower stators. The coil windings are located in the positions, in which they do not interfere with the translational motions of the levitated rotor. The PMs are located on both sides of the center stator and magnetized toward each other to provide PM-biased flux in all 16 airgaps. Fig. 8 shows the simulated stiffness (red lines) and actuation forces (blue lines) in all six DOFs under varying NdFeB35 PM sizes. Note that the performance parameters in the x and y directions, and also in the θ_x and θ_y degrees of freedom, exhibit differences. These differences are attributed to asymmetric magnetic flux leakage paths formed between the rotor and the H-shaped upper and lower stators. Also note that the stiffness in the z direction is near zero since the negative stiffness caused by the upper and lower stators is canceled out against the positive stiffness generated between the center stator and the rotor. Although increasing the PM size enhances the force and torque constants up to a certain level, excessively large magnets cause magnetic saturation in the rotor and stator cores, leading to stiffness saturation and a reduction in actuation forces. This phenomenon degrades both the multi-DOF decoupling performance and the levitation stability. Consequently, the PM dimensions are selected as $50 \times 35\,\text{mm}^2$ with a height of 6 mm, corresponding to a volume of 10.5 cm^3.

Fig. 9 shows the simulated open-loop performances under varying airgap lengths in both the lateral and vertical directions. Note that the lateral airgap length primarily influences the in-plane DOFs (X, Y, and θ_z), while the vertical airgap length predominantly affects the out-of-plane DOFs (θ_x, θ_y, and Z). A smaller airgap generates larger forces and torques but also increases the negative stiffness, leading to reduced system stability. Conversely, an excessively large airgap causes magnetic flux leakage and weakens overall force and torque generation level. This limits the acceleration of the rotor for high-throughput alignment. As a result, the lateral airgap is set to 1.5 mm, emphasizing a high force constant for fast actuation within a sufficient motion range, whereas the vertical airgap is set to 4.5 mm, prioritizing small negative stiffness to ensure stable levitation, as the out-of-plane DOFs function as magnetic bearings rather than alignment actuators.

Finally, the open-loop performance of our actuator prototype is listed in Table II. These values represent the performance of a semi-optimized prototype intended to demonstrate the feasibility of the proposed concept. For example, variations in critical parameters such as airgap lengths or magnet dimensions would

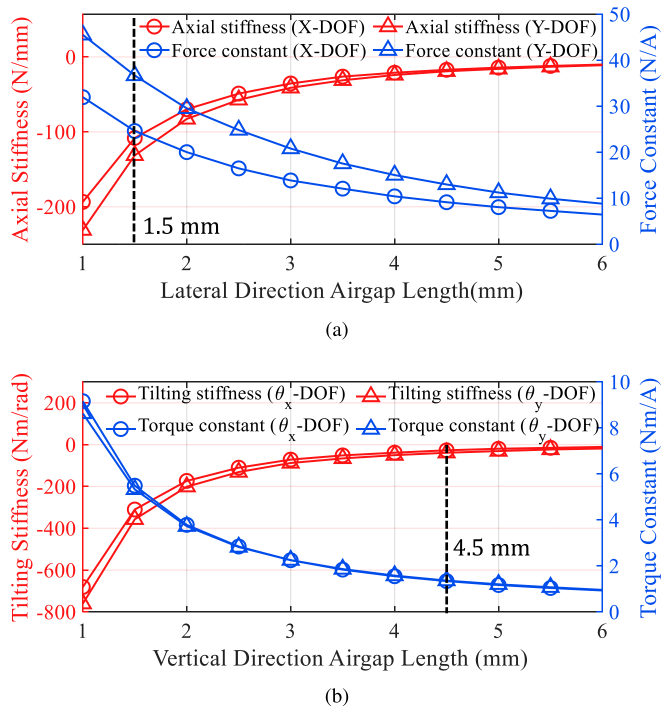


Fig. 9. Simulated stiffness (red lines) and actuation force (blue lines) characteristics with varying airgap lengths in: (a) lateral direction; and (b) vertical direction. The lateral and vertical airgap lengths are selected as 1.5 and 4.5 mm, respectively, as indicated by black dashed lines.

TABLE II
OPEN-LOOP PERFORMANCE ESTIMATED FROM FEA SIMULATIONS

DOF	Performance	Value	Performance	Value
X	Axial stiffness (N/mm)	-109.7	Force constant (N/A)	24.4
Y		-134.6		35.8
Z		0.8		11.8
θ_x	Tilting stiffness (Nm/rad)	-25.9	Torque constant (Nm/A)	1.34
θ_y		-38.0		1.38
θ_z		-28.8		0.97

affect the open-loop performance and would primarily require controller re-tuning, as discussed later in Section IV-A.

B. Position Sensors

We use 12 reflective-type optical sensors (QRE1113 from On Semiconductor, Inc.) to measure the six-DOF rotor positions. These optical sensors are selected for our prototype magnetic levitator module, considering their small size, low cost, and noninterference characteristics with magnetic fields. External voltage sources and resistors are connected to the optical sensors on customized printed circuit boards (PCBs), which are mounted on the aluminum base attached to the optical breadboard, as shown in Fig. 10. In this configuration, four sensors are placed beneath the levitated rotor to measure vertical displacements (z, θ_x, and θ_y), and eight sensors are positioned between the rotor and the center stator to measure lateral displacements (x, y, and θ_z). The sensor locations are selected

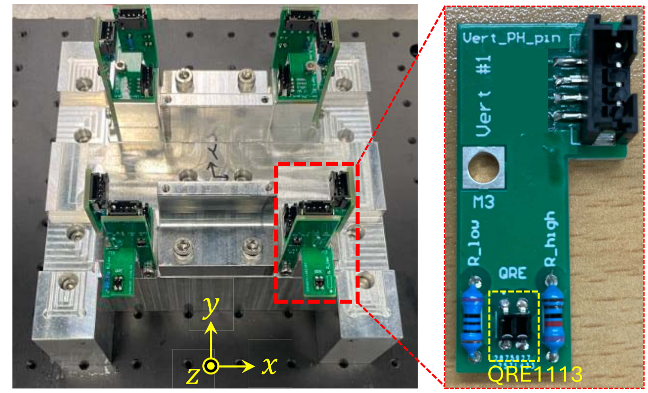


Fig. 10. Photograph of the optical reflective sensing modules mounted on the aluminum base for six-DOF position estimation.

such that the differential signals from multiple sensors enable geometrical decoupling of the six-DOF displacements of the rotor, minimizing cross-coupling among the measured axes. The analog voltage outputs from the sensor modules are sampled at 250 kHz using a 16-bit resolution field-programmable gate array (FPGA) I/O module (PXIe-7846R) with an input range of ± 10 V. To reduce measurement noise, the sensor signals are oversampled in the FPGA module with every ten samples averaged into one measurement in the control loop running at 10 kHz. For calibration, the rotor is displaced relative to the sensing modules using six-DOF microstages. The sensor resolutions are measured to be 151, 86, and 367 nm for the translational displacements of x, y, and z, and 497, 779, and 440 μdeg for the angular rotations of θ_x, θ_y, and θ_z, respectively.

IV. CONTROL SYSTEM

The six-DOF displacements of the levitated rotor are estimated from the optical sensors and fed back for active stabilization. The errors in six DOFs are amplified by the feedback controllers, and the feedback control effort signal is calculated. In addition, the feedforward control effort signal is calculated based on the reference trajectory, which is specifically determined by the measured optical coupling efficiency. The control signals are then combined and distributed to the currents for each coil winding in the stator modules through customized, gain-calibrated current-controlled power amplifiers with proportional–integral regulators. Each amplifier is fine-tuned using high-precision current probe measurement to ensure accurate current delivery across all coils.

A. Active Stabilization

The rotor displacements in all six DOFs except the Z-DOF are open-loop unstable as shown with the negative stiffness in Table II. To stabilize and actively control the six-DOF rotor position, SISO controllers are implemented independently for each DOF in the form of a lead-lag compensator written as

$$C(s) = K_p \cdot \frac{T_i s + 1}{T_i s} \cdot \frac{\alpha \tau s + 1}{\tau s + 1} \tag{1}$$

TABLE III
CONTROLLER PARAMETERS AND LOOP PERFORMANCE

DOF	Controller Parameters				Loop Performance	
	K_p	T_i [ms]	τ [ms]	α	w_c [Hz]	Φ_{pm} [deg]
X	3.4 A/mm	5.3	0.6	10	91	33
Y	3.8 A/mm	5.3	0.6	10	86	29
Z	1.5 A/mm	31.8	1.7	10	32	46
θ_x	2.2 A/deg	26.5	0.8	10	66	34
θ_y	1.4 A/deg	39.8	1.0	10	51	39
θ_z	9.0 A/deg	26.5	0.8	10	58	33

where K_p, T_i, τ, and α denote the proportional gain, the time constant of the lag compensator, the time constant of the lead compensator, and the pole-to-zero ratio in the lead compensator, respectively. Table III provides the controller parameters and the measured loop performance for all six DOFs. The controllers are implemented on a real-time NI-8840 controller, operating at a deterministic sampling rate of 10 kHz. The loop performance measurement indicates that the crossover frequencies w_c for the DOFs required for axial alignment (X and Y) exceed 85 Hz, with phase margins Φ_{pm} greater than 25°. Such stability margins provide sufficient robustness against discrepancies with the predicted performance that may arise during the prototyping process. In addition, model-based feedforward controllers are implemented for the translational DOFs (X, Y, and Z) in parallel with the feedback controllers. The rotor mass and stiffness are preemptively compensated for enhanced servo tracking performance, which is critical for optical fiber alignment. As a result, the translational driving speed of the proposed system reaches up to 80 mm/s, which is about eight times higher than that of commercially available hexapod-piezo stack stages used for fiber alignment [31]. In addition, the average rms current densities at the nominal levitation position are 0.29 A/mm^2 for the center stator and 3.23 A/mm^2 for the lower and upper stators, primarily due to gravity compensation.

B. Multi-DOF Decoupled Control Performance Evaluation

To evaluate the decoupled multi-DOF control performance required for stable levitation and optical fiber alignment, the levitated rotor is commanded to follow an in-plane circular trajectory for 10 s in the X and Y DOFs at 1 Hz with a 100 μm amplitude and 90° phase difference. Fig. 11 summarizes multi-DOF actuation performance of the proposed levitator with six-DOF position data sampled at 5 kHz. Specifically, Fig. 11(a) shows the measured trajectory (blue solid line) against the reference line (red solid line), demonstrating precise and highly repeatable tracking as well as the significant decoupling. Fig. 11(b) and (c) presents the corresponding position and angular errors in the remaining DOFs. The rms errors are measured to be 304, 157, and 399 nm for the translational displacements of x, y, and z, and 799, 990, and 667 μdeg for the angular rotations of θ_x, θ_y, and θ_z, respectively. These results confirm that the proposed levitator maintains stable levitation at its nominal position with minimal cross-coupling, even during in-plane motion generation for optical fiber alignment. Also

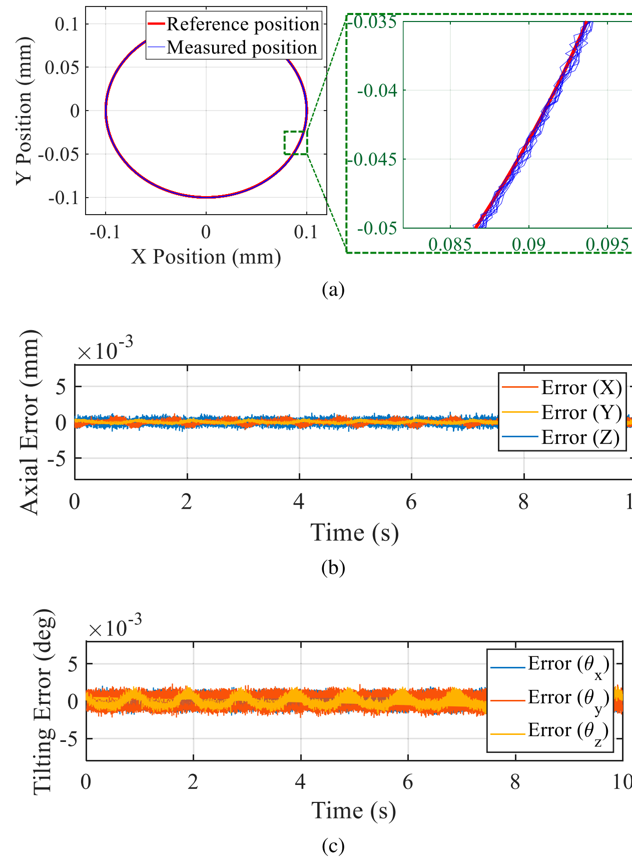


Fig. 11. Decoupled multi-DOF control performance validation with the levitated rotor tracking an in-plane circular trajectory. (a) Reference and measured positions in X and Y DOFs, (b) position errors, and (c) angular errors, showing significantly small errors in all six DOFs. Note that the figure points are plotted at 5 kHz without any additional post-processing.

note that even after hours of continuous operation, no degradation in precision is observed, meaning that coil heating causes negligible performance deterioration over extended actuation.

C. Conventional Spiral Scanning for Rough Alignment

Fig. 12(a) presents the normalized optical coupling efficiency measurement and the axial misalignment identification utilizing the conventional spiral scanning process. During the process, the rotor follows a spiral trajectory and covers 500 μm diameter region, which is 50 times larger than the 10 μm core diameter of the single-mode optical fiber used in the experiment, thereby validating the actuator capability to cover sufficiently large axial offsets for optical fiber alignment. The resulting optical signal exhibits a Gaussian beam distribution with the axial offset between two different single-mode optical fibers of approximately $-30\,\mu$m in the X-DOF and $-60\,\mu$m in the Y-DOF in our experimental setup. Fig. 12(b) illustrates the corresponding time-domain displacements of the rotor during the scanning process. The rotor follows the spiral trajectory for 10 s, making 50 rotations around its nominal center position. During the scanning process, the rms errors are 2.8 and 1.7 μm in the X and Y DOFs, respectively. For the remaining DOFs, the position errors remain well within the sensor resolution

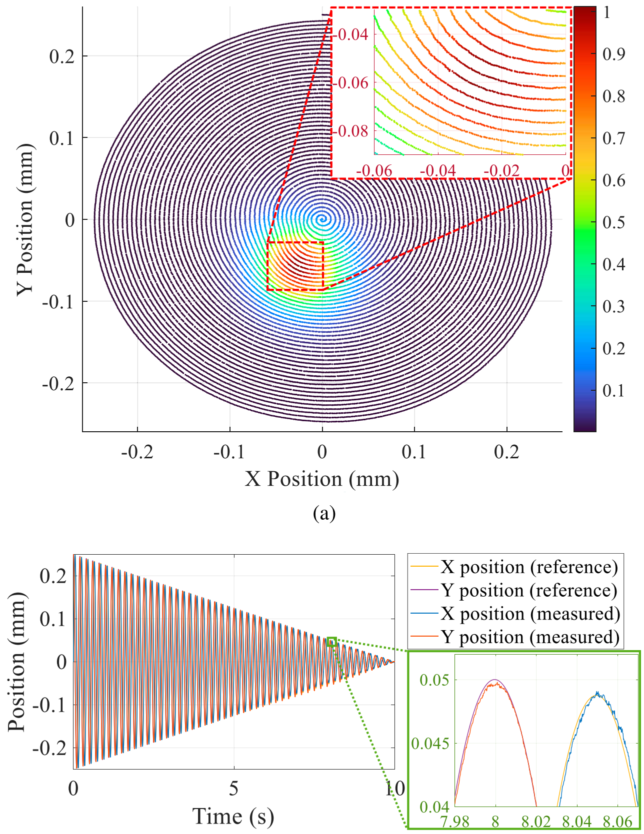


Fig. 12. Experimental areal scanning result sampled at 5 kHz with conventional spiral motion. (a) Optical coupling efficiency measurement, normalized as shown in the color bar, and the axial misalignment identification. After the spiral scanning, the measured optical power distribution roughly reveals an axial offset of -30 and -60 μm in the X and Y DOFs, respectively. (b) Corresponding rotor displacements during the spiral scanning process. Zoomed-in figure highlights the trajectory tracking performance of the levitator module during the oscillatory motion for the spiral trajectory.

level, demonstrating the effective decoupling performance of the system. Specifically, the positioning errors are 502 nm in the Z-DOF and 735, 944, and 492 μdeg for the angular rotations of θ_x, θ_y, and θ_z, respectively. In conventional optical fiber alignment processes, the spiral scanning is typically used as a rough alignment method to detect the first light, followed by fine alignment algorithms, such as hill climbing. In contrast, the proposed method integrates the separated rough and fine alignment steps into a unified process. This allows for rapid and precise alignment, fully leveraging the potential of our magnetic levitator module. The following section introduces the proposed method and discusses the experimental performance in depth.

V. Dynamic Spiral Search for Fully Automated High-Throughput Alignment

This section presents a novel DSS method designed for fully automated high-throughput optical fiber alignment. The performance of this method is compared to the conventional hill climbing method presented in [14]. In addition, we experimentally validate the actuator capability to find the optimal alignment position from various arbitrary fiber misalignment cases across a wide range, which is critical for fully automated robust single-mode optical fiber alignment.

A. Adaptive Amplitude and Center Position Modulation

As shown in Fig. 12(b), the conventional spiral trajectory requires the rotor to oscillate at a constant frequency while the amplitude gradually decreases linearly until it converges to zero. During this process, the center of the spiral motion remains at the nominal position, resulting in fixed scanning pattern regardless of the measured coupling signal $V_{\text{opt}}(t)$ at time t. However, we propose a novel dynamic spiral search method, which introduces two major rules. First, the spiral amplitude is adaptively modulated by the coupling signal, such that the spiral trajectory shrinks faster when a strong optical signal is detected. The reference spiral trajectories $\mathbf{r}_s(t)$ in the X and Y DOFs are then redefined as

$$\mathbf{r}_s(t) = \frac{A}{T}\left(T - (1 + V_{\text{opt}}(t))^A t\right)\begin{bmatrix}\sin(2\pi f t)\\ \cos(2\pi f t)\end{bmatrix} \tag{2}$$

where A, T, and f denote the scanning amplitude, the maximum allowable scanning time, and the scanning frequency, respectively. In this formulation, the key modification lies in the signal-dependent adaptive term $(1 + V_{\text{opt}}(t))^A$, which dynamically accelerates the shrinkage of the spiral path depending on the coupling signal strength. Note that A in the exponential term controls the rate of the shrinkage, allowing fast alignment even with large scanning range. Without this adaptive term, i.e., if $(1 + V_{\text{opt}}(t))^A$ is set to unity, the equation reduces to a conventional spiral trajectory with linear decay over time. The addition of this adaptive term enables the trajectory: 1) to shrink faster when the signal is strong, which indicates proximity to the optimal alignment point; and 2) to decay more slowly when the signal is weak, allowing more time for exploration.

In addition to the adaptive modulation of the spiral path, the proposed DSS method dynamically updates the center of the spiral trajectory in real-time based on the optical feedback signal. The center displacement vector $\mathbf{r}_c(t)$ is updated by

$$\mathbf{r}_c(t + \Delta t) = \mathbf{r}_c(t) + \alpha(t)\mathbf{r}_s(t) \tag{3}$$

where $\alpha(t)$ is a gain parameter that determines the extent to which the current spiral direction influences the update of the center position. This recursive update is performed only when the current optical signal is greater than that of the previous time step. In other words, the spiral center shifts forward only when the coupling signal is increasing, effectively allowing the scan to move toward the region of stronger coupling even when the absolute value of the feedback signal is weak. Specifically, the gain parameter $\alpha(t)$ is written as

$$\alpha(t) = \left(\frac{V_{\text{opt}}(t)}{1 + V_{\text{opt}}(t)}\right)\left(1 - \frac{t}{T}\right)^2 \tag{4}$$

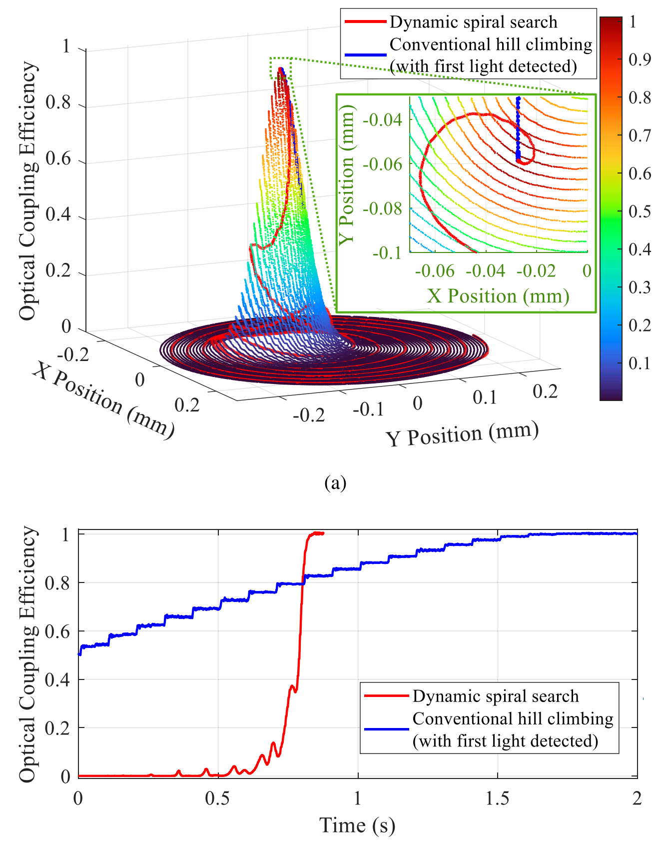
for multiple reasons. First, the term $V_{\text{opt}}(t)/(1 + V_{\text{opt}}(t))$ is initially near-zero with small feedback signal, making the algorithm robust in the region with low signal-to-noise ratio (SNR). This prevents the algorithm from taking wrong steps driven by electric noise. On the other hand, as the feedback signal V_{opt} increases and the SNR improves, the algorithm takes more aggressive steps with certainty, identifying a meaningful Gaussian beam signal. As a result, additive noise in the feedback signal is attenuated through this characteristic before the center is updated in (3), making the DSS method more robust to the small fluctuations caused by the noise in the feedback signal, which is reported to be problematic in conventional feedback-based alignment algorithms [32]. The gain parameter is also modified based on the time t normalized to the total alignment time T. At the beginning of the alignment process, the center can take larger steps for a faster movement to the optimal point. As time approaches the end of the alignment process, the change in the center position is minimized for better positioning stability. The DSS method exhibits a combined effect of adaptation to both the optical signal and the normalized time, while the latter adaptation is squared to put more weight on stability. Finally, the reference trajectory for the alignment $\mathbf{p}(t)$ is defined as

$$\mathbf{p}(t) = \mathbf{r}_s(t) + \mathbf{r}_c(t) \tag{5}$$

which is the superposition of the spiral trajectory $\mathbf{r}_s(t)$ and the center displacement $\mathbf{r}_c(t)$. The DSS algorithm requires only a very small number of custom parameters, yet still generates adaptive motion. This allows robust alignment without repeated parameter tuning for different conditions. Moreover, it remains a simple but practical algorithm that can effectively leverage the strengths of the proposed levitator platform.

B. Alignment Performance of Proposed Method in Comparison With Conventional Algorithm

Fig. 13(a) shows the experimentally validated optical fiber alignment performance of the proposed DSS method compared to the conventional hill climbing algorithm. The alignment trajectories of each method are overlapped on the 3-D view of the optical coupling efficiency measurement from Fig. 12(a). We can see from Fig. 13(b) that the conventional hill climbing trajectory (blue curve) finishes at the optimal position, achieving maximum optical coupling efficiency, after about 19 steps starting from the point where the optical coupling efficiency is about 0.5. Note that such a starting point is selected to make sure for the conventional hill climbing algorithm to converge. The proposed DSS method, on the other hand, starts its trajectory (red curve) from the same position as the conventional spiral trajectory, where there is no optical signal feedback. It however converges at the same optimal position as shown in Fig. 13. During the searching process, the rms tracking errors are 6.4 and 2.9 μm in the X and Y DOFs, respectively. After convergence, the position deviation of the moving fiber from the identified optimal point remains within 760 nm, confirming the sub-micrometer positioning stability of the system. Note that the magnified view in Fig. 13(a) emphasizes that the identified optimal point is located between two neighboring spiral lines, which implies the need for the fine alignment process.



(a)

(b)

Fig. 13. Experimental comparison on the axial optical fiber alignment performance. (a) Alignment trajectories of dynamic spiral search (red curve) and conventional hill climbing algorithm (blue curve) overlaid on the 3-D optical coupling efficiency measurement (color bar) through a conventional spiral trajectory. (b) Time-domain plot of the optical coupling efficiency showing significantly faster alignment by the proposed DSS method.

Fig. 13(b) illustrates the time change of the optical coupling efficiency measurement. The proposed DSS method confirms its alignment accuracy, achieving the correct maximum optical coupling while reducing the alignment time more than twice, namely from 2 s to about 0.84 s. Note that the conventional hill climbing time of 2 s is only for the fine alignment of the conventional two-stage alignment strategy, and therefore the total alignment time should be much slower considering the coarse alignment such as shown in Fig. 12. The slow convergence of the conventional hill climbing method comes from the serial step motions in each DOF and the time required to wait for the complete settling after each step. Unlike the step motions, the DSS trajectory changes dynamically, even making local peaks. This is because the DSS method is made up of the spiral path and the change in the center positions. Even if the change in the center position always leads to an increase in the optical coupling efficiency, the repetitive circular rotations might move further away from the optimal point. However, these rotations decay to zero, and the information acquired through that path is utilized to adaptively update the center

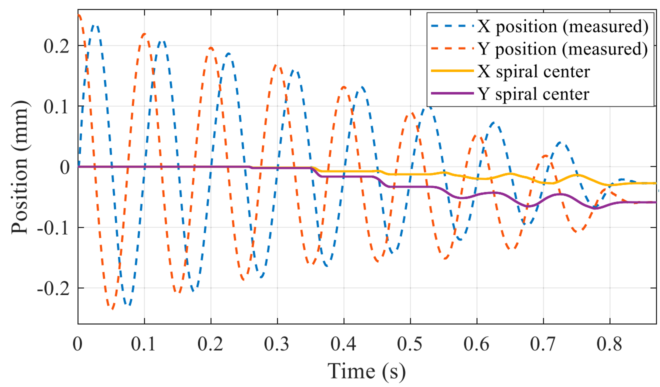


Fig. 14. Measured rotor position during the optical fiber alignment with DSS. Dashed curves represent the rotor position in X and Y DOFs, while solid lines show the corresponding center position, which converges to the optimal point of -27.2 and $-58.8\,\mu$m in X and Y DOFs, respectively.

position, thereby eventually leading to the optimal position at a much faster speed.

Fig. 14 shows the measured position of the rotor in X and Y DOFs for the axial alignment using the proposed DSS method. Dashed lines represent the trajectory that follows the adaptive spiral shape with changes in the center position (solid lines). At the beginning of the process up to about 0.35 s, the center position remains around the nominal position since the optical feedback signal from the Gaussian beam is not detected yet. After that, however, the spiral center is notably attracted toward the optimal position as the spiral path brushes through the side of the Gaussian hill. Starting from about 0.55 s, the spiral center starts to fluctuate. This indicates that the spiral center is now located near the optimal point, and the spiral path is now climbing up to achieve the peak value. At such stage, the center position update slows down and is fine-tuned following the update logic for the adaptive gain as in (4).

The overall validation results manifest that by applying the proposed DSS method to our magnetic levitator, the first light search and fine alignment can be integrated into one single process to achieve high-throughput alignment in a compact footprint. Note that it might pose a significant limitation to apply the proposed DSS method to conventional optical fiber alignment systems with stacked actuators since such architecture lacks the ability to perform seamless rapid alignment with high precision in a unified control process. Such drawback is overcome in our system by integrating high-speed long-range actuation with sub-micrometer precision using a compact six-DOF magnetic levitator platform. Also note that during DSS-based alignment, the rms current density of the center stator coils increases to 0.57 A/mm^2, while that of the lower and upper stator coils remains unchanged. As a result, the total power consumption during the alignment process is approximately 5.7 W. Because the heat generation under this condition is sufficiently low, the actuator maintains thermal stability, which ensures repeatability and long lifetime even under repeated operation in vacuum environments.

C. Large-Range Alignment From Arbitrary Misalignments

To further validate the effectiveness and consistency of the proposed DSS method, we conduct extensive alignment experiments starting from eight random fiber misalignment positions. This scenario reflects realistic conditions in automated systems, where the initial misalignment may vary widely to more than a hundred micrometers. Fig. 15 presents the alignment results from eight distinct experiments, visualized on the measured optical coupling efficiency surface. In these experiments, the optical power sensor signals are low-pass filtered with a cutoff frequency of 200 Hz. This cutoff is chosen to attenuate high-frequency broadband noise of the optical power sensor, including electronic and white noise contributions that are irrelevant to the alignment dynamics, while keeping the phase delay small enough not to compromise the tracking of the meaningful signal dynamics induced by the rotor motion under the DSS algorithm, whose frequency band lies more than a decade below the cutoff. In each experiment, the DSS trajectory (red) starts from a fixed initial position at $x = 0\,\mathrm{mm}$ and $y = 0.25\,\mathrm{mm}$ (same as the conventional spiral trajectory) while the hill climbing algorithm trajectory (blue) starts at the center of the x-y plane. Note that the hill climbing method failed in identifying the maximum point in Fig. 15(g) and (h), due to a significantly small initial coupling signal, while the DSS method successfully and rapidly finds the optimal alignment location in all cases. Specifically, for the cases in which the hill climbing method succeeds, the optimal positions identified by both methods are essentially identical, with differences only at the sub-micrometer level corresponding to the sensor resolution.

Table IV organizes the experimental alignment results quantitatively. The fiber misalignment distance ranges from 9.8 to 150.1 μm. The DSS method consistently completes the alignment in all cases within a relatively consistent time. In contrast, the conventional hill climbing method requires a significantly more time up to 6.20 s, even in successful cases. The alignment time with the conventional method decreases when the fiber misalignment distance is small as in case 6; however, it is still much slower than the proposed DSS method. Furthermore, it fails to converge in cases 7 and 8, indicating vulnerability to large alignment offsets with no notable initial optical coupling signal. By contrast, the DSS algorithm consistently achieved reliable alignment as long as the Gaussian hill is enclosed within the outermost spiral trajectory, which corresponds to offset of about 200 μm in our experimental setup. Such results highlight the robustness of the DSS method across a wide range of initial conditions. The overall results validate that our six-DOF magnetic levitator with the DSS algorithm enables rapid SMF alignment without compromising alignment accuracy, while also manifesting that the high-speed and high-precision actuation of the proposed magnetic levitator can be utilized in practical automated optical fiber alignment applications where the initial misalignment can be significant and unpredictable. In terms of practical deployment, robust DSS algorithm based on the compact and modular structure of the levitator enables straightforward integration into existing photonic packaging lines.

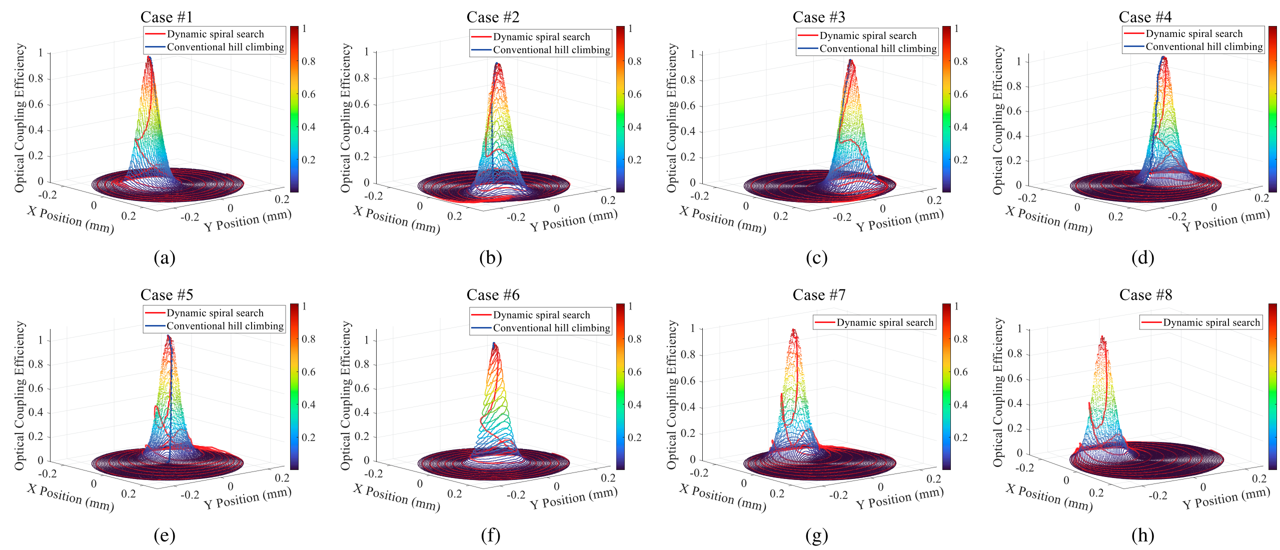


Fig. 15. Axial alignment results in eight random fiber misalignments. Each plot shows the DSS trajectory (red) superimposed on the measured optical signal map. Despite varying the misalignment locations, the proposed DSS method successfully guides the rotor to the same optimal coupling position, within a significantly reduced time as compared to the conventional hill climbing algorithm (blue) in all cases.

TABLE IV
ALIGNMENT RESULTS FROM ARBITRARY FIBER MISALIGNMENTS

Case	Misalignment [μm]	Hill Climbing Time [s]	DSS Time [s]
1	64.7	3.93	0.861 (-78.1%)
2	113.5	5.64	0.844 (-85.0%)
3	107.1	4.98	0.856 (-82.8%)
4	100.0	4.88	0.844 (-82.7%)
5	118.7	6.20	0.842 (-86.4%)
6	9.8	1.66	0.845 (-49.1%)
7	147.2	Failure	0.846
8	150.1	Failure	0.848

VI. CONCLUSION

In this work, we present an active six-DOF magnetic levitator combined with the DSS algorithm for fully automated and high-throughput single-mode optical fiber alignment. The proposed system integrates large-range motion capability, high-speed operation, and sub-micrometer precision within a compact footprint. To validate the alignment performance, the stabilization and servo controllers are implemented for all six DOFs, achieving fast and precise control performance with sub-micrometer tracking errors. The proposed DSS algorithm is applied to our six-DOF magnetic levitator to overcome the limitations of conventional methods by adaptively modulating both the spiral amplitude and center location based on real-time optical feedback. Through a series of experiments, we validate that the DSS method achieves high-speed and reliable alignment without requiring first-light detection, even from various arbitrary fiber misalignments, showing also a significant alignment time reduction of 77.4% on average, while retaining the simple structure and real-time capability of the conventional hill climbing method. To the best of our knowledge, this work represents the first demonstration of applying magnetic levitation to optical fiber alignment. The proposed method can also serve as an efficient automation solution in more complex photonic packaging processes, which require alignment of single-mode components with Gaussian coupling profiles. Furthermore, in practical deployment, the system would also benefit from integration with standard vibration isolation and thermal management systems, as commonly employed in precision photonic assembly, to ensure robust operation in noisy industrial environments. To further investigate the alignment capability of the proposed six-DOF magnetic levitator, we are researching, as our future work, on the alignment of multimode fibers or fiber arrays, in which the full DOF alignment including the tilting motion controls becomes critical. Note that such extensions will also involve utilizing multimodal feedback signals with multiple local peaks.

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